```
    Capstone Project
      ✓ Success on adding grade for a capstone course. (52ms)
      ✓ Success on adding grades for 2 different capstone courses. (104ms)
      ✓ Success on adding grade for a capstone course twice. (100ms)
      ✓ Failure on adding grades for invalid capstone course. (96ms)
      ✓ Failure on adding grades for invalid student. (185ms)
  Events and Eligibility
    ✓ Success on capturing emitted event preRequisiteSatisified. (154ms)
    ✓ Success on capturing emitted event coreCoursesSatisfied. (225ms)
    ✓ Success on capturing emitted event domainRequirementSatisfied. (100ms)
    ✓ Success on capturing emitted event projectRequirementSatisfied. (107ms)
    ✓ Success on capturing emitted event GPARequirementSatisfied. (526ms)
    ✓ Success on capturing emitted event GPA. (485ms)
    ✓ Success on capturing emitted event GPA even if GPA < 2.5. (466ms)
    ✓ Success on eligibility criteria. (447ms)
    ✓ Failure on capturing emitted event preRequisiteSatisified with partial data. (158ms)
    ✓ Failure on capturing emitted event coreCoursesSatisfied with partial data. (251ms)
    ✓ Failure on capturing emitted event domainRequirementSatisfied with no data. (49ms)
    ✓ Failure on capturing emitted event projectRequirementSatisfied with no data. (42ms)
    ✓ Failure on capturing emitted event GPARequirementSatisfied with GPA lower than 2.5. (
ms)
    ✓ Failure on eligibility criteria. (254ms)
  Complete runs
    ✓ Success on run with 3 concurrent users. (1461ms)


 38 passing (12s)
```

Figure 11.8 Output for autotesting DCC.sol

Recall from chapter 10 that the messages for the positive test begin with `'Success on'` and the output messages for negative tests begin with `'Failure on'`. Review the extensive DCCTest.js provided in the test directory, and use it as a model for your future test scripts. The script includes individual tests as well as a complete run testing the entire flow through the smart contract. It is good practice to test the smart contract before coding the web application and UI. Refer to chapter 10 for more details about writing the test script.

11.8 Developing the web application

Now that you have the smart contract coded and tested, you are ready to develop the DCC-app. Review chapters 6-10 for details on developing the web application part of the Dapp. The Dapp